

Vivek Matta

vmatta2002@gmail.com • (781) 800-3636 • vivekmatta.com • github.com/vivekmatta • linkedin.com/in/vivek-matta

EDUCATION

Northwestern University • M.S., Computer Engineering **Sep 2025 - Jun 2026**

- Specialization: Internet of Things & Edge Computing
- Relevant Coursework: Microprocessor Systems, Design and Verification with FPGAs, Electronic System Design I & II
- GPA: 4.0/4.0

Purdue University • B.S., Computer Engineering, Minor in Mathematics **Aug 2021 – May 2025**

- Relevant Coursework: Embedded Systems, Digital Systems and Design, Advanced C Programming
- Awards and Honors: Dean's List (Fall 2021)

WORK EXPERIENCE

Electrical Engineer II • Raytheon **Aug 2026 – Present**

- Observed electrical engineers troubleshooting analog, digital, RF, and power hardware to develop familiarity with laboratory and production test workflows
- Reviewed schematics, engineering drawings, and test procedures alongside engineers during hardware investigations and laboratory activities

Embedded Software Engineer Intern • Cambridge Mobile Telematics **Jun 2025 – Aug 2025**

- Co-developed a benchtop shake table for accelerometer and IoT hardware validation, implementing Embedded C firmware to generate deterministic 0–3200 Hz vibration profiles using hardware timers
- Designed a Python alerting system using SQL queries over historical geolocation data to identify and warn users upon entering high-risk driving zones

Software Engineer Intern • Cambridge Mobile Telematics **Jun 2024 – Aug 2024**

- Integrated Twilio and ConnectedU APIs to process ~5 GB/day of LTE data from 12,000+ IoT SIM cards and built monitoring alerts that reduced response time to usage issues by ~50%

PROJECT EXPERIENCE

Embedded Webcam PCB Design • Electrical Engineering Project, Northwestern University **Jan 2026 – Jun 2026**

- Designed and routed a compact 2-layer PCB in KiCad, integrating camera interfaces, multi-rail power regulation, programming headers, and debug circuitry under mechanical constraints
- Created custom symbols and footprints from datasheets, validated ERC/DRC, generated production-ready Gerbers, and designed an Onshape enclosure for mechanical fit

FM Stereo Radio (SystemVerilog RTL & UVM) • Northwestern University **Jan 2026 – March 2026**

- Designed a fully pipelined FM stereo receiver in SystemVerilog, translating a C reference model into modular RTL with FIFO-based dataflow
- Developed a UVM verification environment and synthesized the design on a Cyclone IV FPGA, analyzing timing, resource utilization, and pipeline bottlenecks

SoCET Project • System-on-Chip Daughter Board Design & PCB Development **Sep 2024 – Dec 2024**

- Designed and routed a compact 2-layer M.2 daughter board integrating power regulation, clock generation, memory, and USB-UART circuitry for SoC bring-up
- Created custom M.2 symbols and footprints from datasheets and validated SPI, UART, and I²S interfaces through schematic review and signal-direction analysis

TECHNICAL SKILLS

Hardware & Embedded: PCB Design (KiCad, Eagle), Embedded C/C++, ESP32, STM32, I²C, SPI, UART, SystemVerilog, FPGA, Oscilloscope, DMM, Logic Analyzer

Tools & Software: Onshape, ModelSim, Synplify Premier, Python, SQL, MATLAB, Git

Hobbies & Interests: Tennis, Pickleball, Trivia, Board Games